

Abimbola R. Ogungbire, Ph.D.

Transportation Engineer & Traffic Safety Researcher

ogungbireabimbola@gmail.com | (336) 405-5918 | Atlanta, GA

linkedin.com/in/abimbola08 | github.com/abimbola08 | [Google Scholar Profile](#)

RESEARCH INTERESTS

Traffic safety analytics; weather-related crash risk prediction and spatiotemporal risk mapping; machine learning and deep learning for transportation systems; crash severity modeling and imbalanced-data treatment; connected and automated vehicles (public perception, agency data readiness); traffic operations, signal performance measures (ATSPM), and microscopic simulation; econometric and discrete-choice modeling of traveler behavior.

EDUCATION

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| Ph.D., Infrastructure and Environmental Systems (Transportation Systems)
University of North Carolina at Charlotte
Dissertation: <i>Novel Machine Learning Techniques for Weather-Related Crash Prediction</i>
Advisor: Dr. Srinivas S. Pulugurtha | Charlotte, NC |
| M.S., Informatics and Analytics (Geospatial Analytics)
University of North Carolina at Greensboro | Greensboro, NC |
| B.Eng., Civil Engineering (Transportation Engineering)
Osun State University | Osogbo, Nigeria |

RESEARCH EXPERIENCE

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| Graduate Research Assistant , University of North Carolina at Charlotte
<i>Infrastructure, Design, Environment and Sustainability (IDEAS) Center, Charlotte, NC</i> | Aug 2021 – May 2023 |
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- Developed a statewide weather-related crash risk workflow for North Carolina using space-time cubes (5-mi grid, monthly aggregation) and equivalent-property-damage-only (EPDO) risk scores, with machine learning models to identify, predict, and interpret high-risk locations.
 - Designed tailored synthetic data generation techniques (SMOTE-N, ADASYN-N extensions) for nominal crash predictors and benchmarked random forest and XGBoost crash severity models; results published in Transportation Research Record.
 - Compared deep learning, machine learning, and statistical models for weather-related crash severity prediction under a Mineta Transportation Institute-funded project (MTI Project 2320); co-authored the final report.
 - Modeled the effect of mountainous terrain on weather-related crash rates using Tobit, latent class Tobit, and two-part truncated log-normal models for western North Carolina.
 - Identified connected and automated vehicle (CAV) data needs of North Carolina public agencies, mapped CAV data to agency use cases, and developed a data readiness framework for NCDOT to prioritize CAV deployments.
 - Built and calibrated VISSIM microsimulation models of the Charlotte Douglas International Airport road network; evaluated unconventional intersection designs (CFI, RCUT, mini roundabouts) and new access routes on travel time reliability (MTI Project 2120).
 - Estimated fixed- and random-parameter ordered logit models of non-motorists' safety perception of autonomous vehicles following a fatal AV crash.

PUBLICATIONS

Peer-Reviewed Journal Articles

- Kalambay, P., Novat, N., Kutela, B., **Ogungbire, A.**, Kitali, A., & Kidando, E. (2026). Autonomous shuttles in the public eye: Mediators on the path to successful expansion. *Case Studies on Transport Policy*, 101719. Elsevier.
- Ogungbire, A.**, & Pulugurtha, S. S. (2025). Spatiotemporal risk mapping of statewide weather-related traffic crashes: A machine learning approach. *Machine Learning with Applications*, 20, 100642. Elsevier.

3. Kalambay, P., Kitali, A., Ngereza, A., Kidando, E., & **Ogungbire, A.** (2025). Autonomous taxis and ride-sharing vehicles: A social construct perspective for future mobility and infrastructure readiness. *Sustainable Cities and Society*, 118, 106060. Elsevier.
4. **Ogungbire, A.**, & Pulugurtha, S. S. (2024). Effectiveness of data imbalance treatment in weather-related crash severity analysis. *Transportation Research Record*, 2678(11), 88–105. SAGE.
5. **Ogungbire, A.**, Kalambay, P., & Pulugurtha, S. S. (2024). Exploring the effect of mountainous terrain on weather-related crashes. *IATSS Research*, 48(2), 136–146. Elsevier.
6. **Ogungbire, A.**, & Pulugurtha, S. S. (2023). Does non-motorists' safety perception of autonomous vehicles vary across opinion change stemming from crash occurrence? Investigating perceptions using fixed and random parameter ordered logit models. *Heliyon*, 9(9), e19913. Elsevier.

Technical Reports

7. **Ogungbire, A.**, Kalambay, P., Gajera, H., & Pulugurtha, S. S. (2023). *Deep learning, machine learning, or statistical models for weather-related crash severity prediction* (MTI Project 2320). Mineta Transportation Institute, San José State University.
8. Pulugurtha, S. S., **Ogungbire, A.**, & Akbari, C. (2023). *Modeling and evaluating alternatives to enhance access to an airport and meet future expansion needs* (MTI Project 2120). Mineta Transportation Institute, San José State University.

Preprints

9. **Ogungbire, A.**, & Pulugurtha, S. (2026). Weather-related crash risk forecasting: A deep learning approach for heterogeneous spatiotemporal data. *arXiv preprint* arXiv:2603.04551.

Doctoral Dissertation

10. **Ogungbire, A. R.** (2024). *Novel machine learning techniques for weather-related crash prediction* [Doctoral dissertation, University of North Carolina at Charlotte].

CONFERENCE PRESENTATIONS

1. **Ogungbire, A.**, & Pulugurtha, S. S. (2026, January). *Spatio-temporal risk mapping of statewide weather-related traffic crashes: A machine learning approach* [Poster presentation, Paper TRBAM-26-05274]. Transportation Research Board 105th Annual Meeting, Washington, DC.
2. Kalambay, P., Novat, N., Kutela, B., **Ogungbire, A.**, Kitali, A., & Kidando, E. (2025, January). *Autonomous shuttles in the public eye: Mediators on the path to successful expansion* [Poster presentation, Paper TRBAM-25-03412]. Transportation Research Board 104th Annual Meeting, Washington, DC.
3. **Ogungbire, A.**, & Pulugurtha, S. S. (2024, January). *The effectiveness of data imbalance treatment in weather-related crash severity analysis* [Lectern presentation, Paper TRBAM-24-06189]. Transportation Research Board 103rd Annual Meeting, Washington, DC.
4. Kalambay, P., **Ogungbire, A.**, & Pulugurtha, S. S. (2024, January). *Vehicle crashes on highway segments with a rail grade crossing: Insights into the key risk factors in North Carolina* [Poster presentation, Paper TRBAM-24-06333]. Transportation Research Board 103rd Annual Meeting, Washington, DC.
5. **Ogungbire, A.**, Diba, D. S., & Pulugurtha, S. S. (2023, January). *How does non-motorists' safety perception of autonomous vehicles vary across opinion change stemming from crash occurrence?* [Lectern presentation, Paper TRBAM-23-03090]. Transportation Research Board 102nd Annual Meeting, Washington, DC.

HONORS, AWARDS & FELLOWSHIPS

ITE Young Leader to Follow , ITE Annual Meeting	2026
One of 20 transportation professionals selected nationally through a competitive application process	
Lifesavers Traffic Safety Scholar , Lifesavers Conference on Roadway Safety	2024
One of 20 students selected nationally through a competitive application process; attended the 42nd Lifesavers Conference, Denver, CO (April 7–9, 2024).	
Excellence in Highway Safety Data Award - 2nd Place , HSIS Student Research Paper Competition	2023
Institute of Transportation Engineers (ITE) and FHWA Highway Safety Information System (HSIS), for “Vehicle Crashes on Highway Segments with a Rail Grade Crossing: Insights into the Key Risk Factors in North Carolina.”	

PROFESSIONAL EXPERIENCE

Traffic Engineer, Gresham Smith | GDOT Traffic Forecasting Nov 2024 – Present
Atlanta, GA

- Review traffic forecasting reports on behalf of the Georgia Department of Transportation Office of Planning and Budgeting.
- Conduct traffic impact analyses for proposed developments and perform roundabout, intersection, and interchange analysis in SIDRA.
- Analyze crash, volume, and roadway data to identify high-risk locations and prioritize safety countermeasures; present spatial crash trends using GIS and Tableau.
- Prepare technical reports and presentations for clients and public agencies.

Traffic Signal & Operations Engineer, Gresham Smith | GDOT SigOps May 2023 – Oct 2024
Atlanta, GA

- Configured, monitored, and retimed signal controllers using Q-Free MaxTime and MaxView to improve arterial flow and reduce delay.
- Extracted and interpreted Automated Traffic Signal Performance Measures (ATSPM) to inform signal timing strategies; built Power BI/DAX dashboards for performance monitoring.
- Simulated signal timing alternatives in VISSIM and TransModeler to quantify impacts on flow, safety, and efficiency.
- Developed automation scripts for data extraction and analysis, substantially reducing manual effort; contributed to technical proposals and business development.

Data Analyst Intern, Volvo Group | IT Jan 2021 – May 2021
New River Valley, VA

- Automated data extraction and preprocessing in R; conducted exploratory analysis and built KPI dashboards for production monitoring.
- Assisted in designing and testing predictive maintenance and anomaly detection models for production line equipment.

Transportation Engineer I, Federal Ministry of Works | Department of Highways Apr 2017 – Jul 2019
Kwara, Nigeria

- Conducted traffic signal warrant analyses and developed signal timing plans from field traffic data; supported signal upgrade implementation and post-installation evaluation.
- Prepared traffic engineering reports covering traffic counts, speed studies, and level-of-service analysis; managed technical proposals with a cumulative value exceeding \$200,000.
- Coordinated with municipal and state agencies to ensure compliance with local traffic engineering standards.

TECHNICAL SKILLS

Modeling & Analysis: Machine learning and deep learning (random forest, XGBoost, ConvLSTM), discrete-choice and count models (ordered logit, Tobit, latent class), time series (ARIMA), spatial statistics and space-time cube analysis, survey and experimental design, mixed-methods research.

Programming & Data: Python, R, DAX, Hadoop, Power BI, Tableau, GIS.

Transportation Software: VISSIM, TransModeler, SIDRA, Q-Free MaxTime / MaxView, ATSPM, Navigator.

Engineering: Microscopic simulation, traffic signal operations and control, operations monitoring, traffic forecasting, traffic impact analysis, systems analysis and evaluation.